

Embedded System & Internet of Things(IoT)

Course By

The IoT Academy

Duration- 45 Days

About Us

We are a fast emerging company imparting quality and affordable programs for colleges and individuals for digital economy skills training, internship and guidance focused on helping people develop the skills they need to thrive in the rapidly growing digital economy. **The IoT Academy** has made a niche name for itself providing rigorous online/offline training in **Data Science, Machine Learning, AI, Python, Java, Internet of Things (IoT), Embedded Systems, Big Data, Data analytics, Industrial IoT, Industry 4.0, Digital marketing, Etc.** Based in Delhi NCR since 2017 we have helped 150+ professionals, 600+ students and 100+ faculties across many states in India.

We have an easy and affordable learning solution that is accessible to millions of learners. With our learners spread across countries like the US, India, UK, Canada, Singapore, Australia, Middle East, Brazil, and many others, we have built a community of over 1 million learners across the globe.



Our Mission

To upskill 5,00,000 young learners to have their career excellence and create Entrepreneurs in next 5 years.



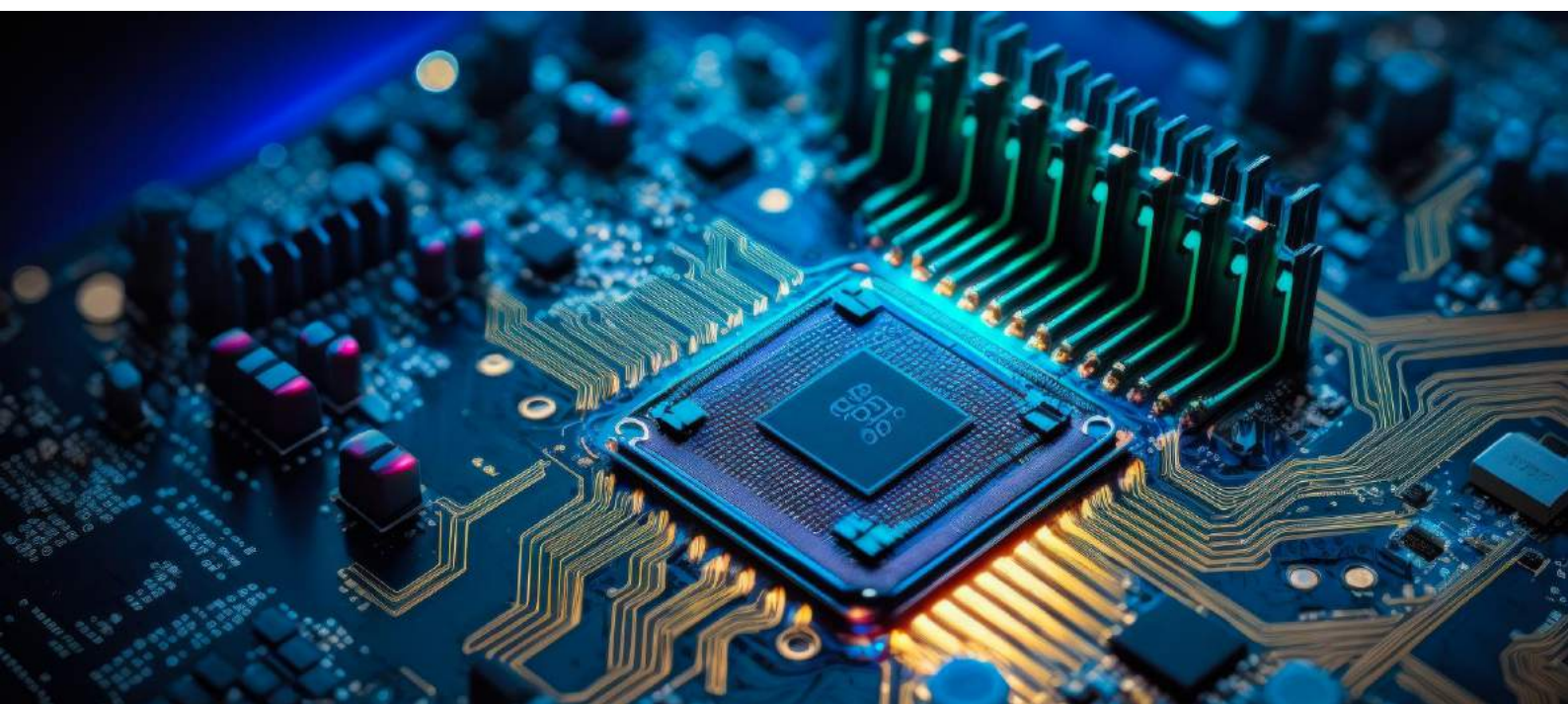
Our Vision

To Impart and contribute significantly in journey of India's 10 Trillion Dollar economy by 2030 by creating high quality Innovations, ambitious and rightly focussed resources.



ABOUT THE PROGRAM

This comprehensive course on Embedded Systems and Internet of Things (IoT) provides students with a deep understanding of the principles, design methodologies, and practical applications of embedded systems and IoT technologies. The course delves into the exciting realm of IoT, exploring the architecture and components that enable devices to connect and communicate over the internet. You will study wireless communication protocols, cloud computing, data analytics, and security mechanisms relevant to IoT applications. Practical projects will guide students in developing IoT-enabled devices, implementing data collection and analysis, and ensuring robust security measures.



Introduction to IoT

- Introduction to Internet of Things (IoT)
- Different skills required to become an IoT developer



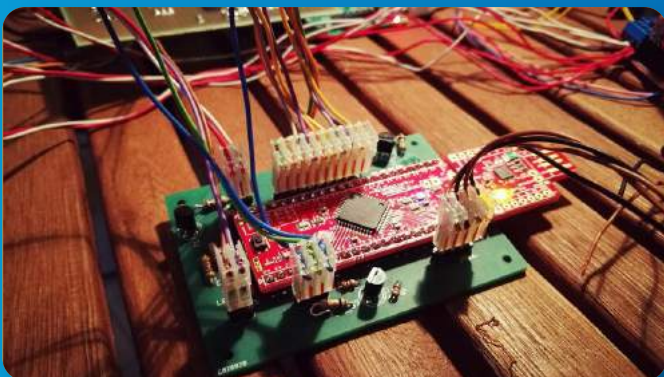
Embedded C

- Why C in embedded system
- Data types and operators
- Selection & Loops
- Functions
- Pointers
- Callbacks functions
- Structure
- Memory and scope



IoT Components

- Sensors
- Switches
- Microcontrollers
- Gateways
- Cloud
- Overview of ML, AI & Data science



Ardiuno Basics

- Creating Sketches

Python Basics

- Python Basic

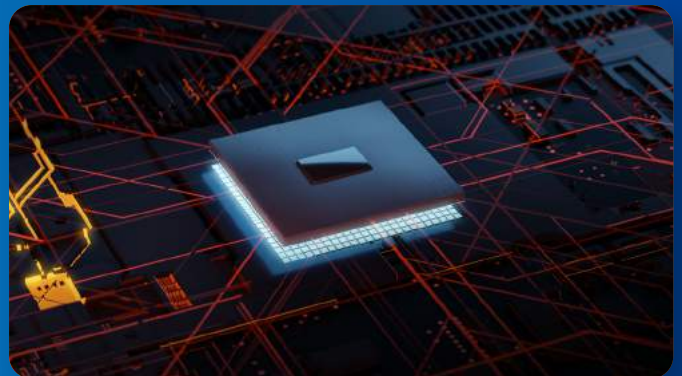


Raspberry Pi

- Controlling Raspberry Pi GPIOs

Embedded Protocols

- UART, I2C, SPI



IoT Protocols

- MQTT, COAP, , TCP/IP, HTTP

ConnectiFY

- Using Node MCU and Arduino with ConnectiFY

PROJECT 1 & 2

Certification:

Upon successfully completing this program, you'll Learn **Internet of Things (IoT)**.

This certificate will testify to your skills as an expert in **Internet of Things**

Certification ID: xxxxxx



CERTIFICATE OF COMPLETION

This Is To Certify That

Mr./Ms. Name Surname

Is Given The Certificate Of Achievement For Completing The

Internet of Things (IoT)

This Certificate Of Completion Is Being Given In Appreciation Of His/Her Commitment And Efforts And Accumulation Of Knowledge & Skills Which Enabled The Course To Be Completed. We Wish Him/Her Success Throughout His/Her Career.

xxxx 2023 - xxxx 2023

Date



Kaushlendra Singh Sisodia
Chief Mentor, The IoT Academy
Director, Uniconverge Technologies

Thank You!!



**The IoT
Academy**

Connecting The Unconnected



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